

Name: _____



New York State Testing Program

Mathematics Test Session 1

Grade 8

Spring 2024

RELEASED QUESTIONS

Session 1



TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.

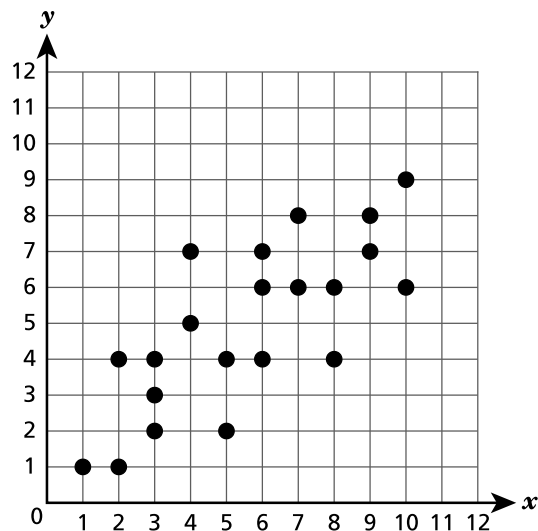
1

A group of friends went to a movie theater and paid \$15.00 for each movie ticket. The group of friends shared one container of popcorn that cost \$8.99. Which equation can be used to determine the total cost, c , for n tickets and the container of popcorn?

- A $c = 15n + 8.99$
- B $c = 8.99n + 15$
- C $c = 15(n + 8.99)$
- D $c = 8.99(n + 15)$

2

A scatter plot is shown below.



Which pair of points could be used to draw a line that best represents the relationship of the data?

- A $(4, 7)$ and $(8, 4)$
- B $(3, 3)$ and $(9, 7)$
- C $(2, 4)$ and $(6, 4)$
- D $(3, 2)$ and $(3, 4)$

GO ON

- 4 Triangle ABC is rotated 90° about the origin and then reflected over the y axis to form triangle $A'B'C'$. One of the angles in triangle ABC has a measure of 115° . One of the angles in triangle $A'B'C'$ has a measure of 40° . Which is the measure of an angle in triangle ABC?

A 25°
B 75°
C 155°
D 180°

- 5 What is the value of x in the equation $\frac{2}{3}x - 7 = 5 - \frac{3}{5}x$?

A $3\frac{18}{19}$
B $9\frac{9}{19}$
C $15\frac{1}{5}$
D $19\frac{1}{5}$

7

Two electricians, Electrician A and Electrician B, offer pricing plans for their work. Each electrician charges an initial fee for a service call plus an hourly rate. The charges for each electrician are represented by the equation and the table shown below.

ELECTRICIAN A

$$C = 25x + 50$$

ELECTRICIAN B

Time (hours)	Total Charge (dollars)
3	130
4	150
5	170

Which statement comparing the charges for each electrician is true?

- A Electrician A has an initial fee and an hourly rate that are both less than those for Electrician B.
- B Electrician A has an initial fee and an hourly rate that are both greater than those for Electrician B.
- C Electrician A has an initial fee that is less than that for Electrician B. The hourly rate for Electrician A is greater than that for Electrician B.
- D Electrician A has an initial fee that is greater than that for Electrician B. The hourly rate for Electrician A is less than that for Electrician B.

GO ON

9

What is the slope of the line on a coordinate plane that passes through the points $(2, 2)$ and $(-1, -2)$?

A $-\frac{4}{3}$

B $-\frac{3}{4}$

C $\frac{3}{4}$

D $\frac{4}{3}$

GO ON

10

A sphere has a radius of 4 inches. A cone has a radius of 3 inches and a height of 8 inches. Which expression represents the difference in volume, in cubic inches, between the sphere and the cone?

A $\pi \left[\frac{4}{3}(4^3) - \frac{1}{3}(3^2)(8) \right]$

B $\pi \left[\frac{4}{3}(4^2) - \frac{1}{3}(3^2)(8) \right]$

C $\pi \left[\frac{4}{3}(4^3) - \frac{1}{3}(8)^2(3) \right]$

D $\pi \left[\frac{4}{3}(4^2) - \frac{1}{3}(8)^2(3) \right]$

GO ON

12

Which value of x makes the equation $x^3 = 64$ true?

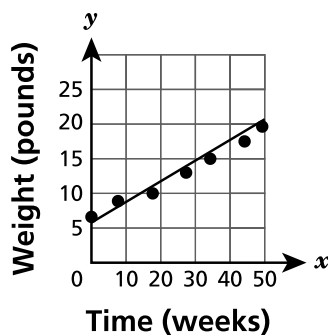
- A 4
- B 8
- C 16
- D 32

GO ON

18

The scatter plot below shows the weight, in pounds, of an alligator as it grows at a zoo. The line $y = 0.3x + 5.8$ has been drawn to **best** represent the relationship between the alligator's weight and growth time.

ALLIGATOR'S WEIGHT



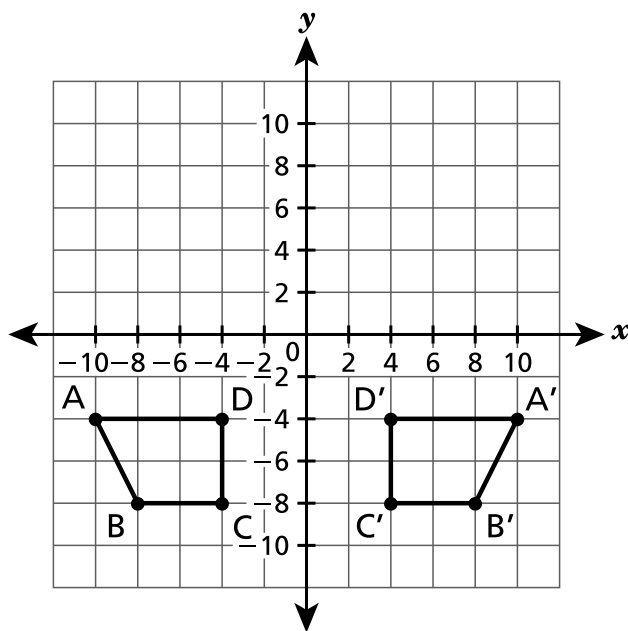
Which statement **best** describes what the number 0.3 in the equation represents in this situation?

- A The starting weight, in pounds, of the alligator.
- B The approximate number of pounds the alligator gains each week.
- C The maximum number of pounds the alligator gains each week.
- D The average number of pounds of food the alligator is fed each week.

GO ON

19

Trapezoid ABCD and its image A'B'C'D' are shown on the coordinate plane below.



Which series of transformations could be applied to map trapezoid ABCD onto trapezoid A'B'C'D'?

- A reflection over the x axis and then a rotation of 180° about the origin
- B reflection over the x axis and then a rotation of 90° about the origin
- C reflection over the y axis and then a rotation of 180° about the origin
- D reflection over the y axis and then a rotation of 90° about the origin

20

Which equation represents a linear function?

- A $y = \frac{1}{2}x - 3$
- B $y = x^2 + 5$
- C $y = x^2 + 2x$
- D $y = \frac{1}{5}x^3$

GO ON

22

On a coordinate plane, triangle ABC is rotated 90 degrees clockwise about the origin and then dilated with a scale factor of 2 centered at the origin to form triangle $A'B'C'$. Which statement describes the relationship between triangle ABC and triangle $A'B'C'$?

- A** They are similar and congruent.
- B** They are similar but not congruent.
- C** They are congruent but not similar.
- D** They are neither congruent nor similar.

23

One of the angles in a triangle measures x degrees. Another angle in the triangle measures y degrees. Which expression represents the measurement, in degrees, of the third angle of the triangle?

- A** $180 - (x + y)$
- B** $180 - x + y$
- C** $x + y - 180$
- D** $x + y + 180$

GO ON

27 Which expression is equivalent to 3^5 ?

A $\frac{3^{10}}{3^5}$

B $\frac{3^{15}}{3^3}$

C $\frac{9^{10}}{3^5}$

D $\frac{9^3}{9^5}$

GO ON

30

Which list of numbers could represent the side lengths of a right triangle?

- A** 5, 10, 13
- B** 5, 12, 17
- C** 10, 24, 26
- D** 10, 24, 68

STOP

Name: _____



New York State Testing Program

Mathematics Test Session 2

Grade 8

Spring 2024

RELEASED QUESTIONS

Session 2



TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.
- Be sure to show your work when asked.
- Be sure to explain your answer when asked.

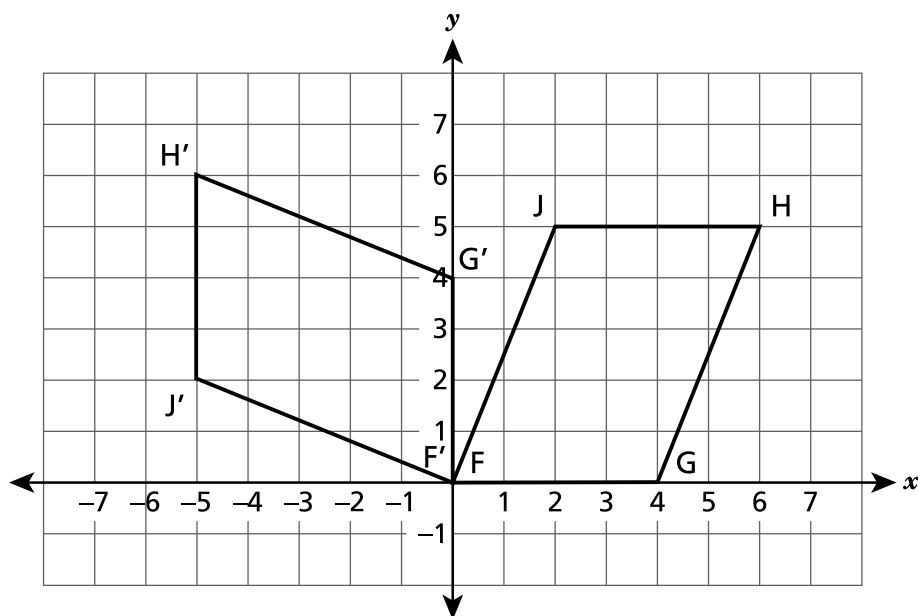
The table below represents a function.

x	y
2.5	7.5
3.5	10.5
4.5	13.5
5.5	16.5

Which statement describes the function?

- A The function is nonlinear because the y intercept is 0.
- B The function is linear because the rate of change is constant.
- C The function is linear because the y intercept is a constant value.
- D The function is nonlinear because the rate of change is not constant.

Quadrilateral $FGHJ$ is rotated about the origin to form quadrilateral $F'G'H'J'$.

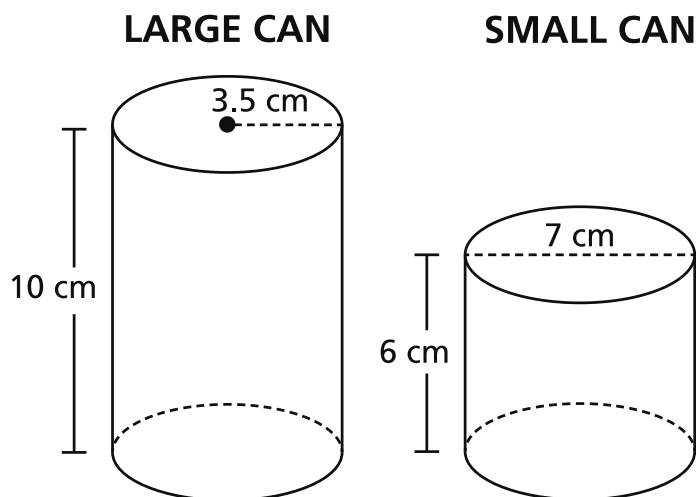


Which two sides are equal in length?

- A $\overline{FG}$ and $\overline{F'J'}$
- B $\overline{JF}$ and $\overline{H'J'}$
- C $\overline{GH}$ and $\overline{H'G'}$
- D $\overline{HG}$ and $\overline{H'J'}$

35

A manufacturer makes cylindrical cans in two sizes. The dimensions of each can are shown in the diagram below.



What is the difference between the volumes, in cubic centimeters, of the large can and the small can in terms of π ?

- A 4π
- B 49π
- C 73.5π
- D 155.5π

36

An equation is shown below.

$$2(3x + 1) = x + 1 + 5x$$

Which statement about the equation is true?

- A It has no solutions.
- B It has exactly one solution.
- C It has exactly two solutions.
- D It has an infinite number of solutions.

GO ON

37 Which statement **best** describes the value of $\sqrt{2}$?

- A** between 0.5 and 1.0
- B** between 1.5 and 2.0
- C** between 1.0 and 1.5 but closer to 1.0
- D** between 1.0 and 1.5 but closer to 1.5

38 A set of ordered pairs is shown below.

$\{(-3, 3), (1, 1), (4, 2), (-1, -1), (-2, -1), (3, 1), (-2, 3)\}$

Which ordered pair should be removed to make the set a function?

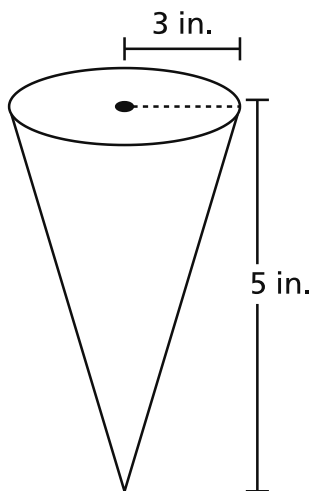
- A** $(-3, 3)$
- B** $(4, 2)$
- C** $(-2, -1)$
- D** $(3, 1)$

39

This question is worth 1 credit.

A movie theater sells popcorn in cone shaped containers as shown below.

POPCORN CONTAINER



What is the volume, in cubic inches, of the popcorn container? Round your answer to the nearest tenth.

Answer _____ cubic inches

GO ON

40

This question is worth 1 credit.

The area of a square shaped garden is 324 square feet. What is the length, in feet, of each side of the garden?

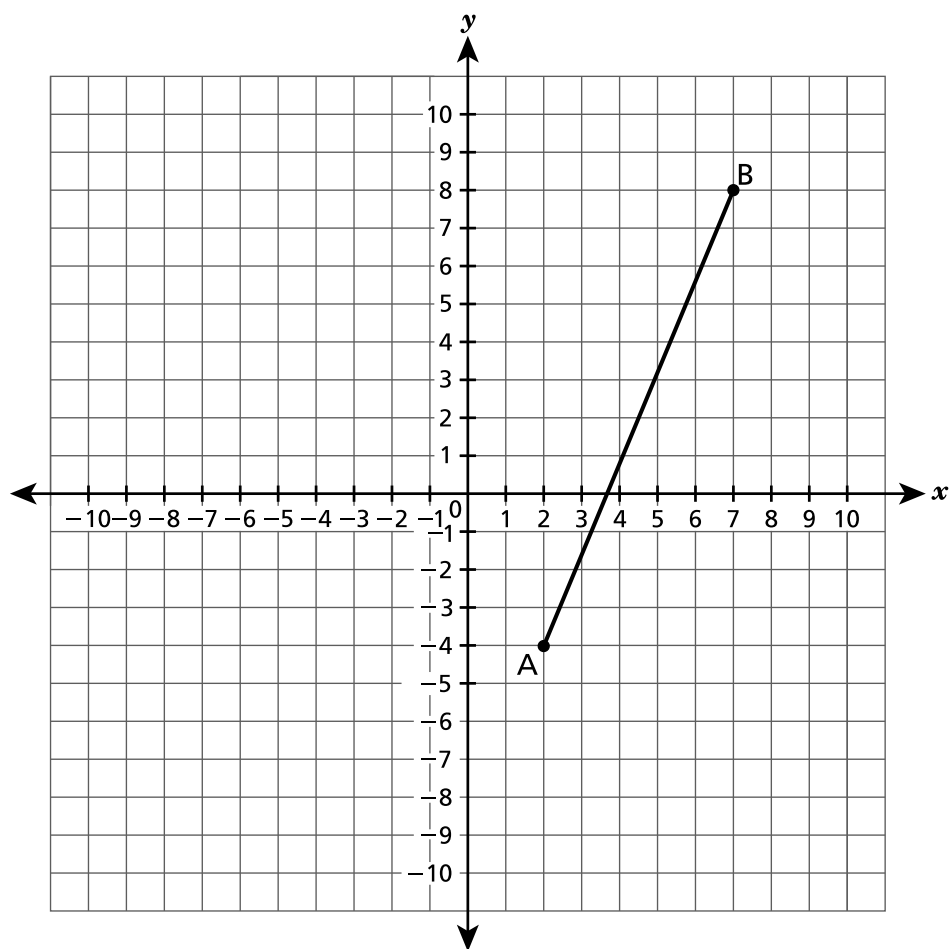
Answer _____ feet

GO ON

41

This question is worth 1 credit.

Line segment AB is graphed on the coordinate plane shown below.



What is the length, in units, of line segment AB ?

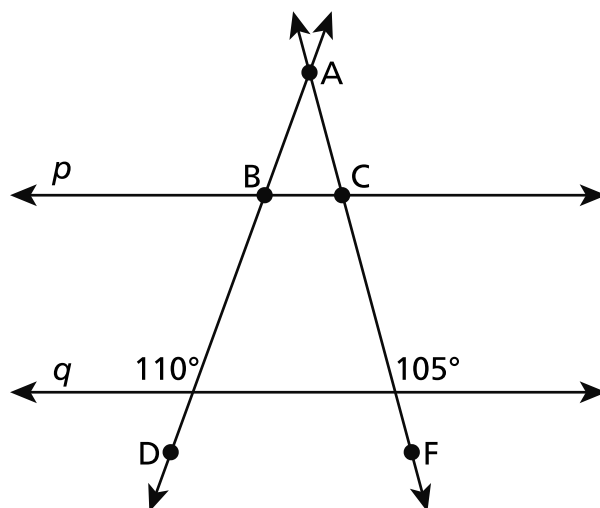
Answer _____ units

GO ON

42

This question is worth 2 credits.

In the figure below, line p is parallel to line q and lines AD and AF are transversals.



What is the measure, in degrees, of $\angle BAC$?

Show your work.

Answer _____ degrees

GO ON

43

This question is worth 2 credits.

The equation $y = 1.5x + 29$ is used to model the yearly salary, y , of an employee, in thousands of dollars, where x is the number of years the employee has worked for the company. What does the slope of the line represent in this situation?

Explain your answer.

GO ON

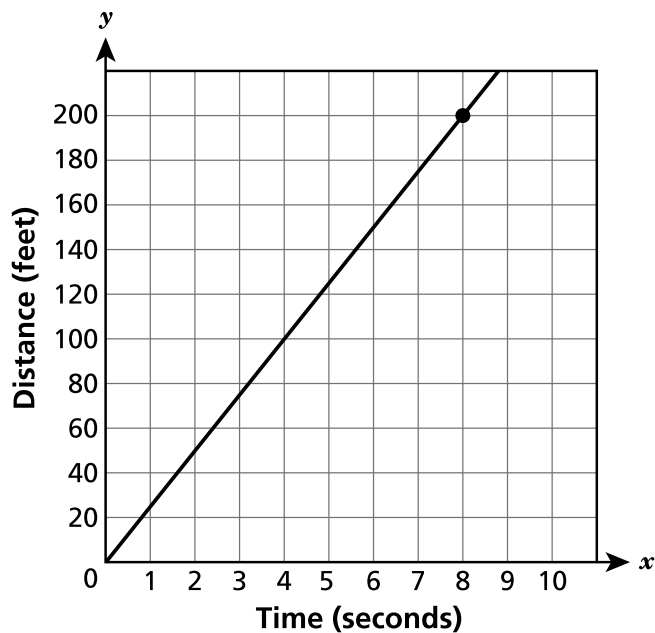
44

This question is worth 2 credits.

A dog owner collected data to see which of his two dogs runs at the greater speed. The graph and the table below show the relationship between the time, in seconds, and the distance, in feet, each dog ran.

DOG A

Time, x (seconds)	Distance, y (feet)
2	56
4	112
6	168
8	224

DOG B

What is the difference, in feet per second, between the speeds of the two dogs?

Show your work.

Answer _____ feet per second

GO ON

45

This question is worth 2 credits.

Two ordered pairs of a linear function are shown below.

$$\left(2, 4\frac{1}{2}\right), \left(3, 5\frac{1}{4}\right)$$

What is the rate of change for the function?

Show your work.

Answer _____

GO ON

46

This question is worth 2 credits.

What value of x makes the equation shown below true?

$$\frac{1}{4}(3x - 8) + 4 = 2(x - 4)$$

Show your work.

Answer $x =$ _____

GO ON

47

This question is worth 2 credits.

A list of numbers is shown below.

- $\sqrt{49}$
- $1.\bar{3}$
- $\sqrt{32}$
- $\frac{7}{2}$
- 1.234

Classify each number as either rational or irrational. Be sure to include how you know a number is rational.

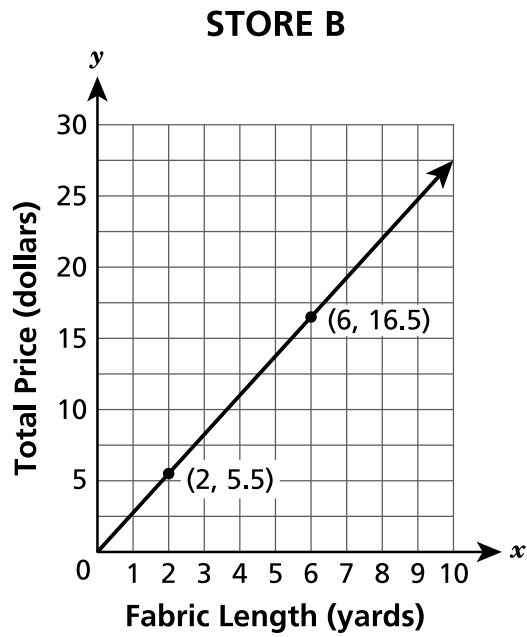
Explain your answer.

GO ON

48

This question is worth 3 credits.

Store A and Store B sell fabric for different prices. The equation $y = 3.5x$ represents the price, y , in dollars, for x yards of fabric at Store A. The graph below represents the price for the same type of fabric at Store B.



What is the unit rate for the price of fabric, per yard, at each store?

Store A \$ _____ per yard of fabric

Store B \$ _____ per yard of fabric

How much more would the price of 9 yards of fabric be at Store A than at Store B?

Show your work.

Answer \$ _____

STOP